

HENDERSON, KY, USA

WMO: 724238

Lat: 37.800N

Lon: 87.683W

Elev: 118

StdP: 99.92

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-10.1	-18.8	0.7	-11.8	-16.3	0.9	-9.6	11.2	7.3	10.4	5.9	2.8	320	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.1	24.7	32.9	24.4	32.2	24.1	26.4	32.6	25.7	31.6	24.9	30.5	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.0	30.4	23.8	18.9	29.6	22.9	17.8	28.4	82.7	32.8	79.5	31.6	76.0	30.5	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.5	DB	-18.4	36.1	5.2	1.3	-22.2	37.0	-25.2	37.8	-28.1	38.5	-31.9	39.4	(4)
(5)			WB	-18.5	27.9	4.7	1.0	-21.8	28.5	-24.5	29.1	-27.1	29.6	-30.5	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	0.7	2.7	7.9	14.2	19.5	23.9	25.3	24.6	20.9	14.6	7.7	2.8
	DBStd	10.02	6.57	6.28	5.96	5.11	4.36	3.03	2.60	2.73	3.96	4.99	5.25	5.64
	HDD10.0	990	297	214	113	19	1	0	0	0	0	13	103	230
	HDD18.3	2455	548	437	328	141	41	2	0	1	20	138	319	481
	CDD10.0	2373	8	11	47	146	294	417	473	451	327	156	35	8
	CDD18.3	796	0	0	3	18	76	169	215	193	97	23	1	0
	CDH23.3	7655	0	1	23	161	668	1616	2109	1861	974	236	6	0
	CDH26.7	3077	0	0	2	27	209	657	914	802	401	65	1	0
Wind	WSAvg	3.1	3.9	3.9	3.9	4.0	3.3	2.6	2.0	1.9	2.2	2.9	3.4	3.6
Precipitation	PrecAvg	1092	89	78	112	112	132	91	96	71	81	86	95	96
	PrecMax	1573	162	160	278	276	296	156	189	231	278	233	169	191
	PrecMin	773	32	23	51	42	31	37	42	17	0	14	12	31
	PrecStd	190	36	42	60	58	66	36	42	50	65	54	41	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	21.3	26.3	28.9	32.6	35.0	37.0	37.0	34.9	31.3	23.9	19.0
		MCWB	14.9	16.1	17.8	19.5	22.6	24.9	26.1	25.7	23.3	22.0	16.6	15.6
	2%	DB	16.9	17.9	22.7	27.4	31.0	33.0	33.9	34.0	32.8	28.1	21.2	16.2
		MCWB	13.8	13.7	15.7	18.4	22.0	24.0	25.7	25.0	22.8	19.6	15.7	12.9
	5%	DB	13.7	15.9	20.8	25.3	28.9	32.3	32.6	32.5	31.3	26.2	18.7	13.7
		MCWB	11.0	12.0	15.0	17.3	20.8	23.7	25.4	24.5	22.3	18.0	14.2	10.8
	10%	DB	10.9	12.6	17.6	22.8	27.5	31.0	31.4	31.2	28.8	23.3	16.6	11.4
		MCWB	8.2	9.2	12.7	16.1	20.0	23.2	24.5	23.9	21.1	17.1	12.2	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.0	18.8	21.1	24.0	26.3	27.9	27.3	25.0	23.1	18.9	16.4
		MCDB	17.7	20.3	24.9	27.3	30.0	32.9	33.6	33.3	32.2	29.6	23.3	18.7
	2%	WB	14.2	14.8	16.8	19.7	22.9	25.4	26.8	26.1	24.0	21.4	16.6	13.7
		MCDB	16.9	17.4	21.9	25.4	29.2	31.6	32.6	32.6	30.9	26.2	19.8	15.9
	5%	WB	11.2	12.4	15.1	18.4	21.8	24.5	26.0	25.3	23.0	19.5	14.6	11.2
		MCDB	13.5	15.4	19.6	23.7	27.7	30.6	31.7	31.3	29.3	24.0	18.0	13.5
	10%	WB	7.8	9.2	13.3	17.0	20.7	23.7	25.2	24.4	21.9	17.9	12.6	8.7
		MCDB	9.8	12.7	17.4	21.9	26.1	29.4	30.5	29.7	27.2	22.7	16.4	10.9
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.3	10.5	11.8	11.0	10.9	11.0	11.8	13.2	12.7	10.5	8.2
		MCDBR	11.1	13.3	13.8	14.0	12.9	12.7	12.5	13.6	15.5	15.0	13.7	10.6
		MCWBR	9.7	10.4	8.8	8.1	6.3	5.9	6.4	6.7	7.1	8.0	9.5	8.9
	5% WB	MCDBR	10.2	12.1	12.0	11.8	11.3	11.0	11.1	12.1	13.2	12.5	11.3	9.7
MCWBR		9.5	10.5	8.4	7.7	6.1	5.6	6.2	6.2	6.5	7.0	9.2	9.0	
Clear-Sky Solar Irradiance	taub	0.317	0.328	0.368	0.406	0.438	0.451	0.478	0.484	0.415	0.363	0.330	0.311	
	taud	2.457	2.402	2.351	2.250	2.212	2.198	2.135	2.106	2.292	2.430	2.469	2.475	
	Ebn at Noon	866	901	892	874	848	835	808	792	832	850	842	847	
	Edh at Noon	83	99	115	135	143	145	153	154	121	95	81	76	
All-Sky Solar Radiation	RadAvg	2.00	2.70	3.75	4.97	5.59	6.37	6.11	5.62	4.67	3.48	2.37	1.72	
	RadStd	0.21	0.32	0.34	0.29	0.48	0.39	0.39	0.29	0.33	0.37	0.26	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+77

Nomenclature: See separate page